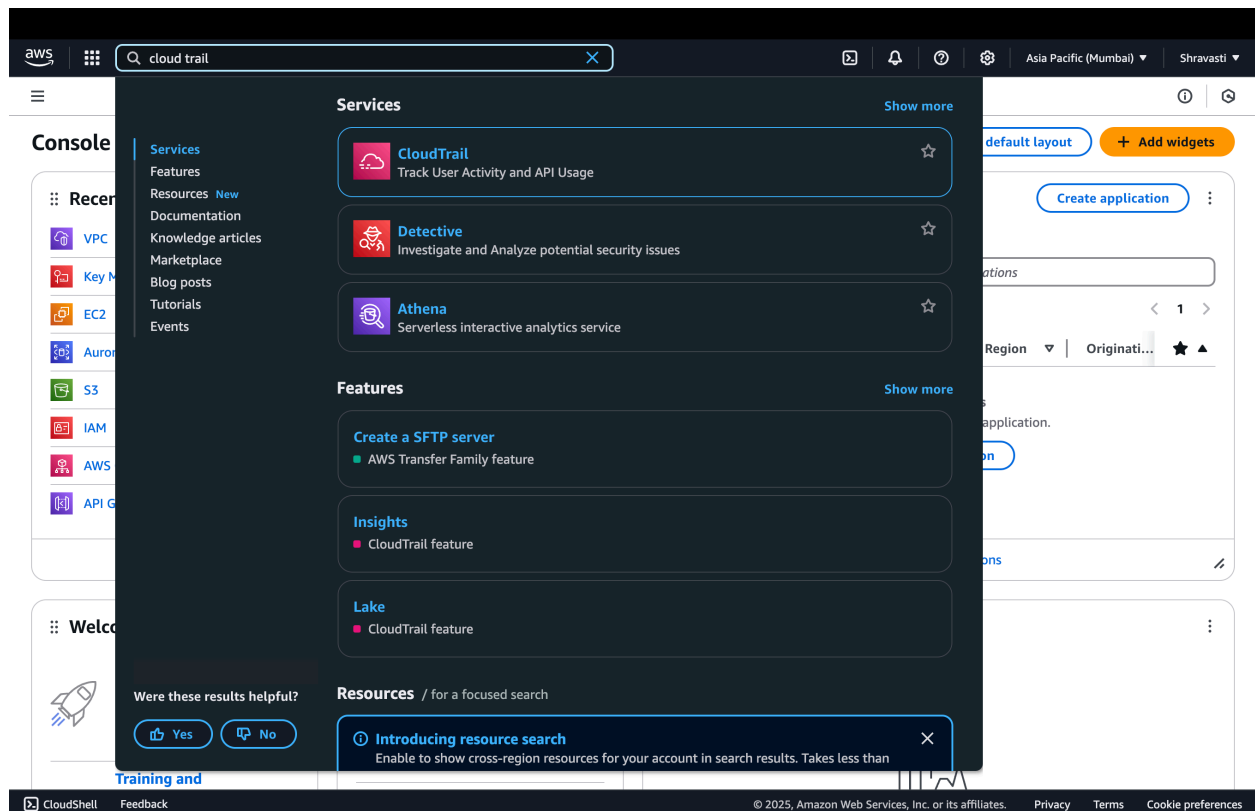


lab 10 - monitoring and alerting with cloud trail and cloud watch

1. Log in to AWS and navigate to Cloud Trail



2. Click on "Create a trail"

aws [Search] [Option+S]

Asia Pacific (Mumbai) Shrivasti

Management & Governance

AWS CloudTrail

Continuously log your AWS account activity

Use CloudTrail to meet your governance, compliance, and auditing needs for your AWS accounts.

Create a trail with AWS CloudTrail

Get started with AWS CloudTrail by creating a trail to log your AWS account activity.

[Create a trail](#)

How it works



Capture

Record activity in AWS services as AWS CloudTrail events



Store

AWS CloudTrail delivers events to the AWS CloudTrail console, Amazon S3 buckets, and optionally Amazon CloudWatch Logs

Pricing

[Pricing](#)

Getting started

[What is AWS CloudTrail?](#)

[How AWS CloudTrail works](#)

[Services that integrate with AWS CloudTrail](#)

More resources

CloudShell Feedback

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3. Name your trail and click on "create a trail"

aws

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[Option+S]

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CloudTrail

> Quick trail create

Quick trail create

Trail details
Start logging management events by creating a trail with simplified settings. Logs are sent to an S3 bucket we create on your behalf. To choose a different bucket or additional events, go to the full [Create trail](#) workflow.

A trail created in the console is a multi-region trail. [Learn more](#)

Trail name
Enter a display name for your trail.

3-128 characters. Only letters, numbers, periods, underscores, and dashes are allowed.

Trail log bucket and folder

Logs will be stored in aws-cloudtrail-logs-619071313125-d85e4972/AWSLogs/619071313125

Though there is no cost to log these events, you incur charges for the S3 bucket that we create to store your logs.

Cancel

Create trail

CloudShell

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4. Trail will be successfully created

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CloudTrail

Trails

Trails

Copy events to Lake

Delete

Create trail

	Name	Home region	Multi-region trail	Insights	Organization trail	S3 bucket	Log file prefix	CloudWatch Logs log group	Status
☐	my-first-trail	Asia Pacific (Mumbai)	Yes	Disabled	No	aws-cloudtrail-logs-619071313125-d85e4972	-	-	Logging

CloudShell

Feedback

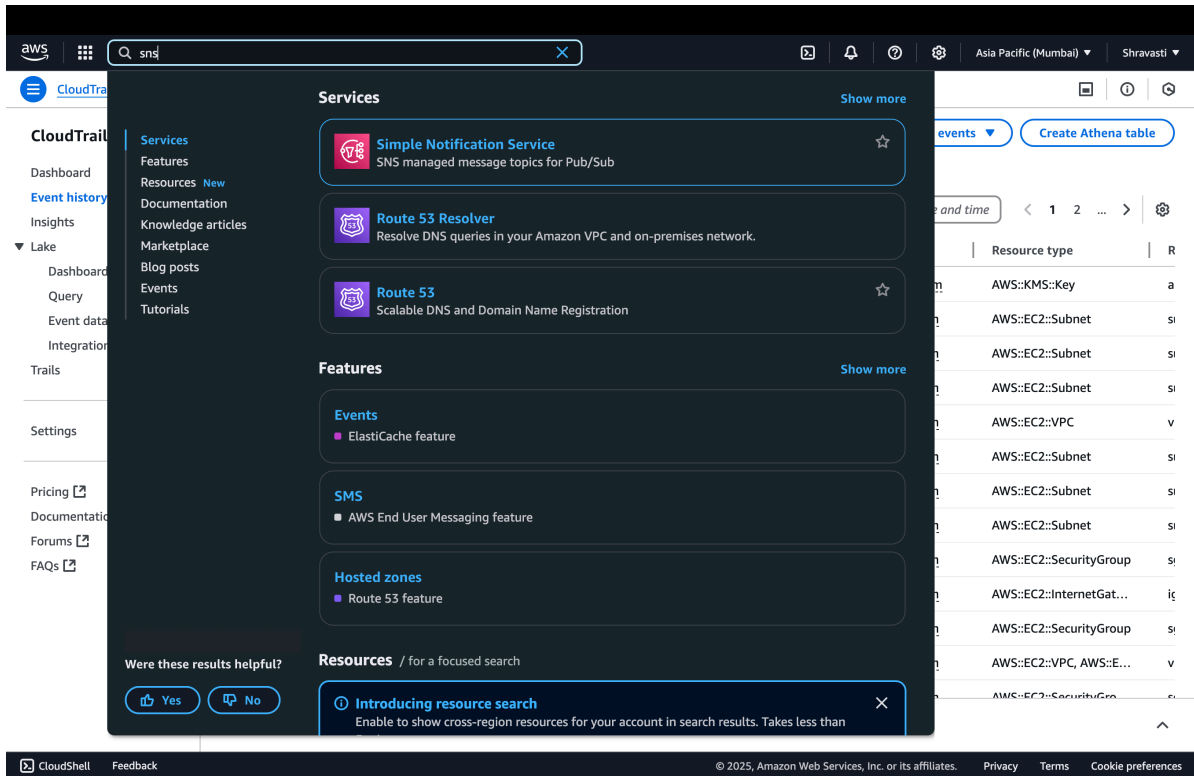
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5. Navigate to "Amazon Simple Notification Service (SNS)"



6. Name your SNS Topic and click on next step

aws

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New Feature

Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Application Integration

Amazon Simple Notification Service

Pub/sub messaging for microservices and serverless applications.

Amazon SNS is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and event-driven serverless applications. Amazon SNS provides topics for high-throughput, push-based, many-to-many messaging.

Create topic

Topic name

A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.

MyFirstSNSTopic

Next step

[Start with an overview](#)

Pricing

Amazon SNS has no upfront costs. You pay based on the number of messages you publish, the number of messages you deliver, and any additional API calls for managing topics and subscriptions. Delivery pricing varies by endpoint type.

[Learn more](#)

Benefits and features

Reliably deliver messages with durability

Amazon SNS uses cross availability zone message storage to provide

Automatically scale your workload

Amazon SNS leverages the proven AWS cloud to dynamically scale

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7. Select type → standard and a display name

aws

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Amazon SNS > Topics > Create topic

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create topic

Details

Type [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

MyFirstSNSTopic

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)

To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

shravsTopic

Maximum 100 characters.

► Encryption - optional

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

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8. Select method → Basic, allow everyone to publish messages to the topic and allow everyone to subscribe to it

[Option+S]

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[Amazon SNS](#) > [Topics](#) > Create topic

► **Encryption - optional**

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

▼ **Access policy - optional** [Info](#)

This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

Choose method

☒ **Basic**
 Use simple criteria to define a basic access policy.

☐ **Advanced**
 Use a JSON object to define an advanced access policy.

Publishers

Specify who can publish messages to the topic.

Everyone
 Anybody can publish

Subscribers

Specify who can subscribe to this topic.

Everyone
 Any AWS account can subscribe to the topic

JSON preview

```

{
  "Version": "2008-10-17",
  "Id": "__default_policy_ID",
  "Statement": [
    {
      "Sid": "__default_statement_ID",
      "Effect": "Allow",
      "Principal": {
        "AWS": "*"
      },
      "Action": [
        "SNS:Publish",
        "SNS:RemovePermission",
        "SNS:SetTopicAttributes",
        "SNS:DeleteTopic",
      ]
    }
  ]
}

```

► **Data protection policy - optional** [Info](#)

This policy defines which sensitive data to monitor and to prevent from being exchanged via your topic.

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9. Create topic

aws

Search

[Option+S]

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Amazon SNS > Topics > Create topic

"Action": [
"SNS:Publish",
"SNS:RemovePermission",
"SNS:SetTopicAttributes",
"SNS:DeleteTopic",

► Data protection policy - optional [Info](#)

This policy defines which sensitive data to monitor and to prevent from being exchanged via your topic.

► Delivery policy (HTTP/S) - optional [Info](#)

The policy defines how Amazon SNS retries failed deliveries to HTTP/S endpoints. To modify the default settings, expand this section.

► Delivery status logging - optional [Info](#)

These settings configure the logging of message delivery status to CloudWatch Logs.

► Tags - optional

A tag is a metadata label that you can assign to an Amazon SNS topic. Each tag consists of a key and an optional value. You can use tags to search and filter your topics and track your costs. [Learn more](#)

► Active tracing - optional [Info](#)

Use AWS X-Ray active tracing for this topic to view its traces and service map in Amazon CloudWatch. Additional costs apply.

Cancel

Create topic

CloudShell

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10. Topic created successfully

aws [Search] [Option+S]

Amazon SNS > Topics > MyFirstSNSTopic

Amazon SNS

- Dashboard
- Topics
- Subscriptions
- ▼ Mobile
 - Push notifications
 - Text messaging (SMS)

MyFirstSNSTopic [Edit] [Delete] [Publish message]

Details

Name MyFirstSNSTopic	Display name shravsTopic
ARN arn:aws:sns:ap-south-1:619071313125:MyFirstSNSTopic	Topic owner 619071313125
Type Standard	

< **Subscriptions** | Access policy | Data protection policy | Delivery policy (HTTP/S) | Delivery status logging >

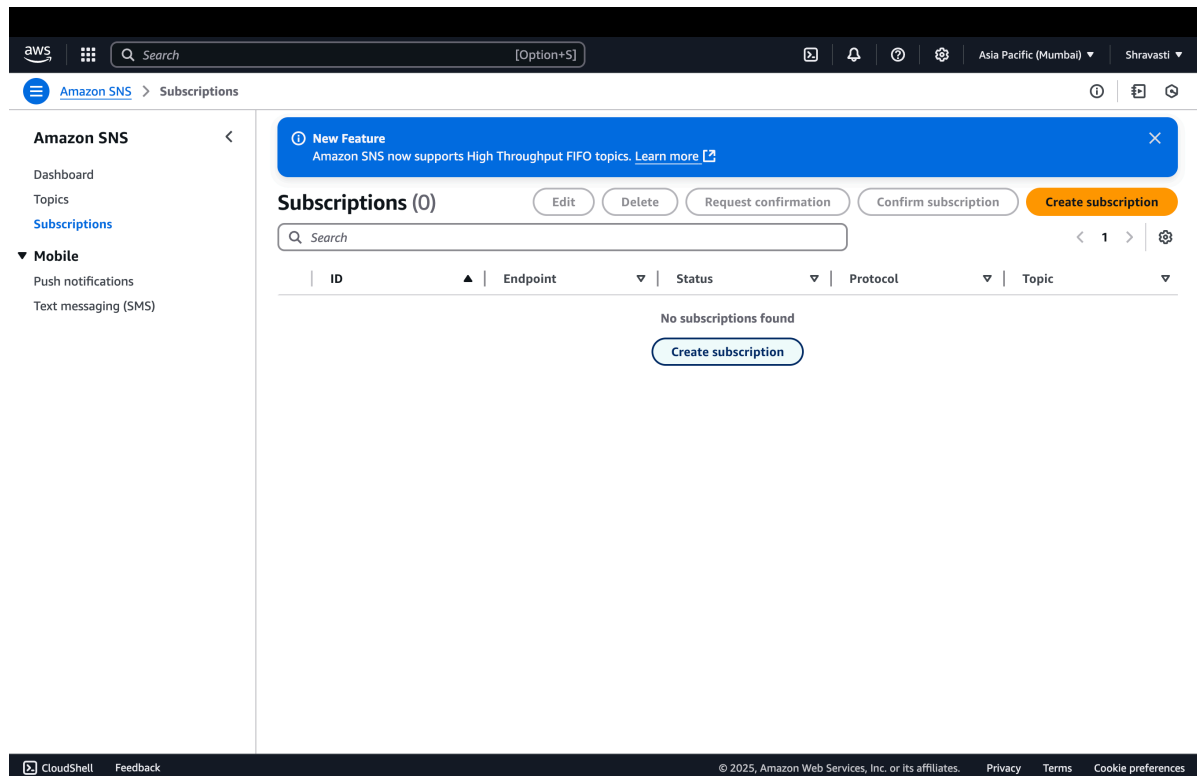
Subscriptions (0) [Edit] [Delete] [Request confirmation] [Confirm subscription] [Create subscription]

[Search]

ID	Endpoint	Status	Protocol
No subscriptions found			

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11. Create a subscription for the topic



12. Select your topic ARN, specify protocol → email, add an email address as endpoint and click on create subscription

aws

Search

[Option+S]

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Shravasti

Amazon SNS > Subscriptions > Create subscription

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Create subscription

Details

Topic ARN

Protocol
The type of endpoint to subscribe

Endpoint
An email address that can receive notifications from Amazon SNS.

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

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13. Subscription will be created successfully

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Amazon SNS > Topics > MyFirstSNSTopic > Subscription: a52306ce-efd2-4454-a6e6-3094905ebd99

Amazon SNS

- Dashboard
- Topics
- Subscriptions

▼ **Mobile**

- Push notifications
- Text messaging (SMS)

Subscription: a52306ce-efd2-4454-a6e6-3094905ebd99 [Edit] [Delete]

Details

ARN arn:aws:sns:ap-south-1:619071313125:MyFirstSNSTopic:a52306ce-efd2-4454-a6e6-3094905ebd99	Status ⌚ Pending confirmation
Endpoint shravasti563@gmail.com	Protocol EMAIL
Topic MyFirstSNSTopic	
Subscription Principal arn:aws:iam::619071313125:root	

Subscription filter policy | Redrive policy (dead-letter queue)

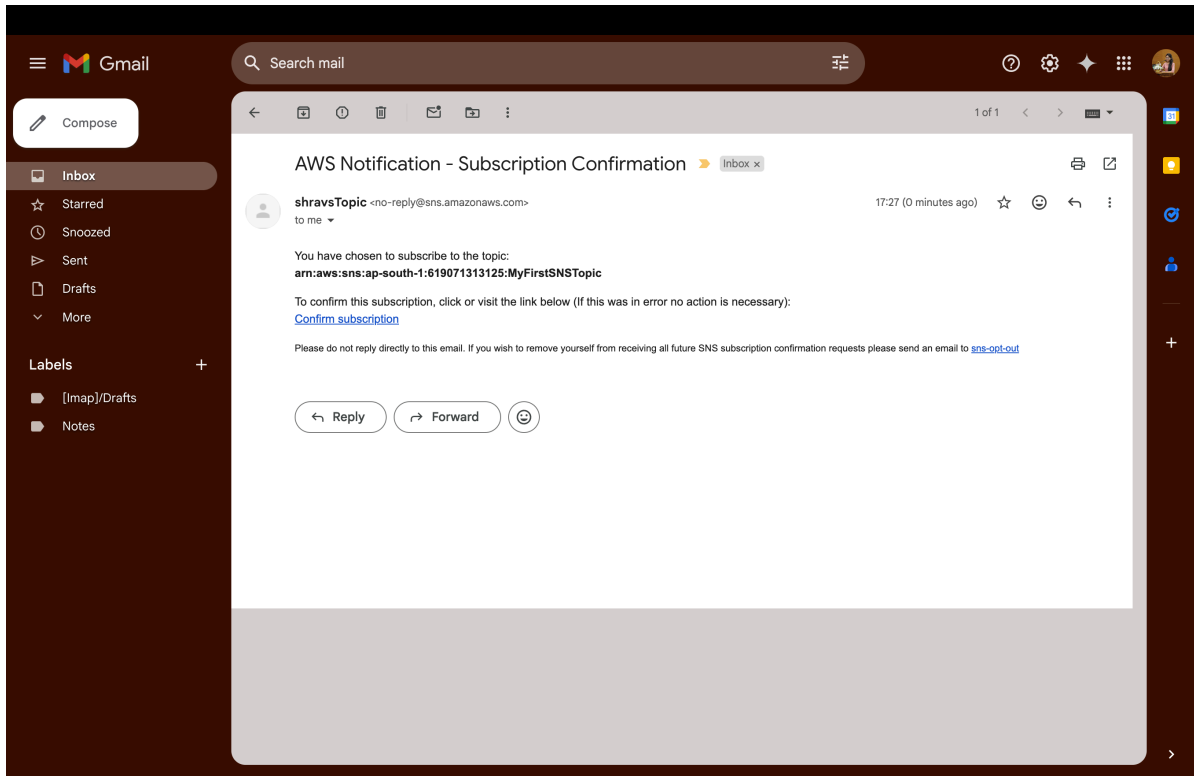
Subscription filter policy [Info](#)

This policy filters the messages that a subscriber receives.

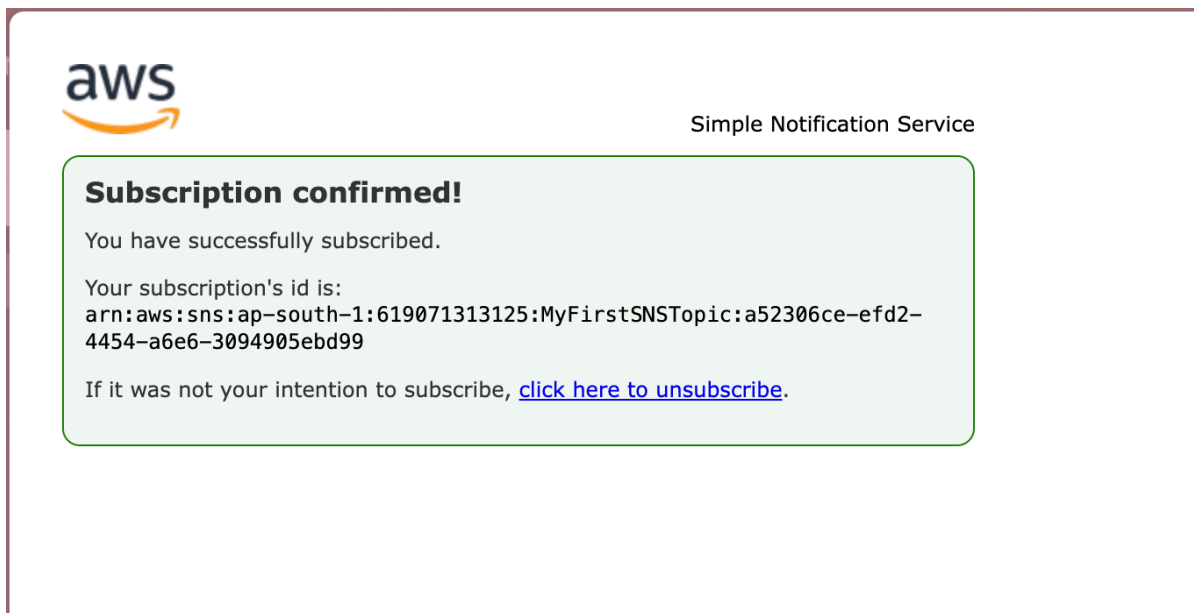
No filter policy configured for this subscription.
To apply a filter policy, edit this subscription.

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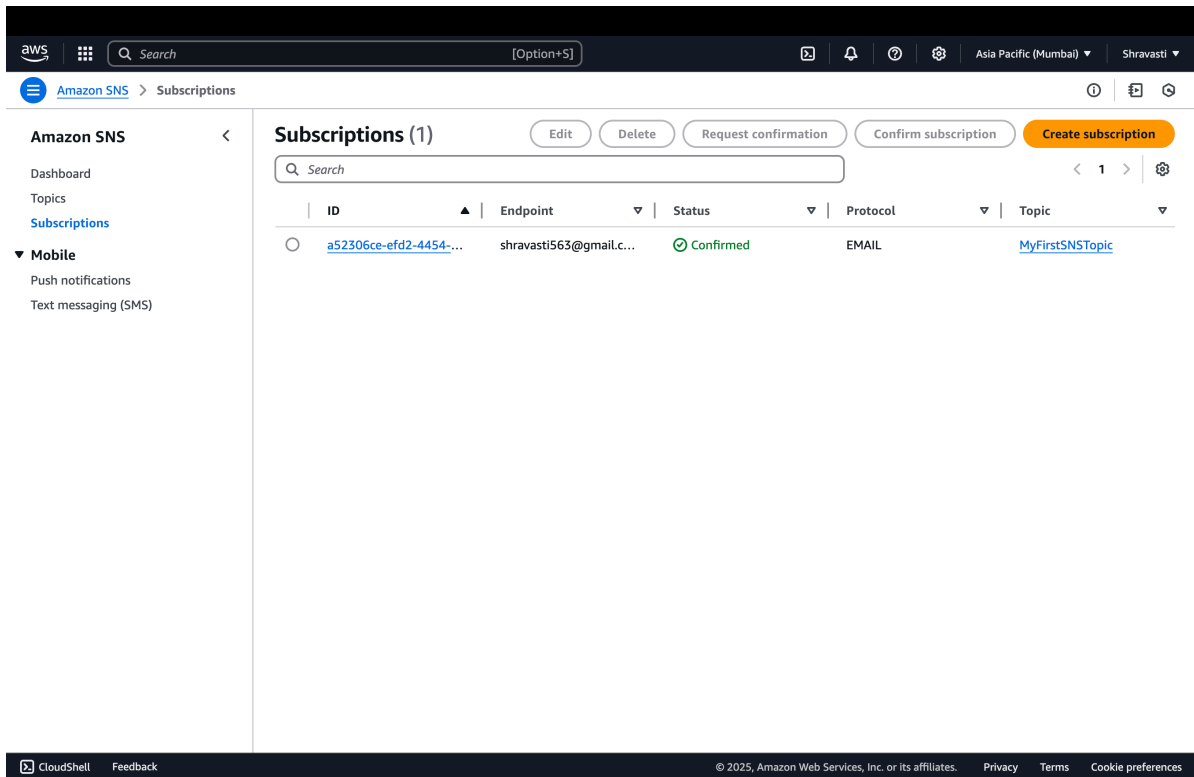
14. You will receive confirmation through email



15. Confirm subscription



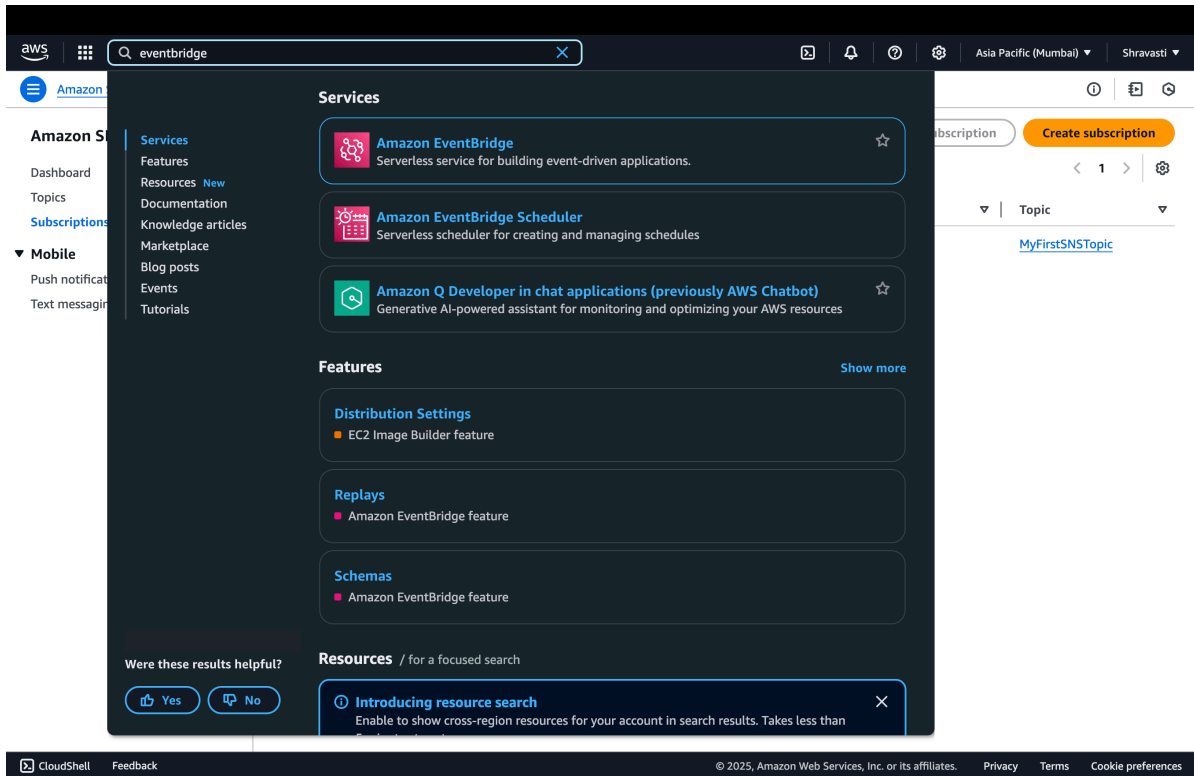
16. Refresh the AWS page



The screenshot shows the Amazon SNS Subscriptions page in the AWS console. The top navigation bar includes the AWS logo, a search bar, and the user's name 'Shravasti'. The left sidebar shows the 'Amazon SNS' menu with options like 'Dashboard', 'Topics', 'Subscriptions', and 'Mobile'. The main content area is titled 'Subscriptions (1)' and contains a table with one subscription. The table has columns for ID, Endpoint, Status, Protocol, and Topic. The subscription is confirmed and uses an email endpoint.

ID	Endpoint	Status	Protocol	Topic
a52306ce-efd2-4454-...	shravasti563@gmail.c...	Confirmed	EMAIL	MyFirstSNSTopic

17. Navigate to Amazon EventBridge



18. Select "EventBridge Rule" and click on "create rule"

aws

Search

[Option+S]

Asia Pacific (Mumbai)

Shravasti

Amazon EventBridge

Dashboard

Developer resources

Buses

Pipes

Scheduler

Integration

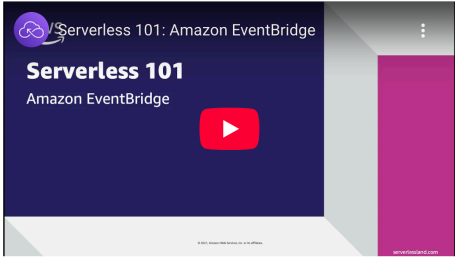
Schema registry

Amazon EventBridge

A serverless service for building event-driven applications

Amazon EventBridge is a serverless service that uses events to connect application components together, making it easier for developers to build scalable event-driven applications.

How it works



Get started

- ☒ **EventBridge Rule**
A rule matches incoming events and sends them to targets for processing.
- ☐ **EventBridge Pipes**
A pipe connects an event source to a target with optional filtering and enrichment.
- ☐ **EventBridge Schedule**
A schedule invokes a target one-time or at regular intervals defined by a cron or rate expression.
- ☐ **EventBridge Schema registry**
Schema registries collect and organize schemas.

Create rule

19. Name the rule, give it a description, and select rule type as “rule with an event pattern”

The screenshot shows the AWS EventBridge console interface. The top navigation bar includes the AWS logo, a search bar, and user information. The left sidebar contains a navigation menu with categories like Developer resources, Buses, Pipes, Scheduler, Integration, and Schema registry. The main content area is titled 'Define rule detail' and shows a progress bar with five steps: Step 1 (Define rule detail), Step 2 (Build event pattern), Step 3 (Select target(s)), Step 4 (optional, Configure tags), and Step 5 (Review and create). The 'Define rule detail' step is active.

Define rule detail [Info](#)

Rule detail

Name

 Maximum of 64 characters consisting of numbers, lower/upper case letters, -, _

Description - optional

Event bus [Info](#)
 Select the event bus this rule applies to, either the default event bus or a custom or partner event bus.

☒ **Enable the rule on the selected event bus**

Rule type [Info](#)

☒ **Rule with an event pattern**
 A rule that runs when an event matches the defined event pattern. EventBridge sends the event to the specified target.

☐ **Schedule**
 A rule that runs on a schedule

[Cancel](#) [Next](#)

20. Select event source → AWS Events, sample event type → AWS Events

The screenshot shows the Amazon EventBridge console interface. The top navigation bar includes the AWS logo, a search bar, and the text "[Option+S]". Below this, the breadcrumb trail reads "Amazon EventBridge > Rules > Create rule". The left sidebar contains a menu with categories: "Developer resources" (Learn, Sandbox, Quick starts), "Buses" (Event buses, Global endpoints, Archives, Replays), "Pipes" (Pipes), "Scheduler" (Schedules, Schedule groups), "Integration" (Partner event sources, API destinations, Connections, Schemas), and "Schema registry" (Schemas). The main content area is titled "Build event pattern" and shows a five-step process: Step 1 (Define rule detail), Step 2 (Build event pattern - selected), Step 3 (Select target(s)), Step 4 - optional (Configure tags), and Step 5 (Review and create). The "Build event pattern" step is expanded, showing options for "Event source" (AWS events or EventBridge partner events, Other, All events) and "Sample event - optional" (AWS events, EventBridge partner events, Enter my own). A "Sample events" section with a filter dropdown is also visible.

21. Event pattern → Custom pattern (JSON editor)

aws [Search] [Option+S] Asia Pacific (Mumbai) Shravasti

Amazon EventBridge > Rules > Create rule

Amazon EventBridge

- Dashboard [New](#)
- ▼ **Developer resources**
 - Learn
 - Sandbox
 - Quick starts
- ▼ **Buses**
 - Event buses
 - [Rules](#)
 - Global endpoints
 - Archives
 - Replays
- ▼ **Pipes**
 - Pipes
- ▼ **Scheduler**
 - Schedules
 - Schedule groups
- ▼ **Integration**
 - Partner event sources
 - API destinations
 - Connections [Updated](#)
- ▼ **Schema registry**
 - Schemas

Event pattern [Info](#)

Creation method

- ☐ Use schema
Use an Amazon EventBridge schema to generate the event pattern.
- ☐ Use pattern form
Use a template provided by EventBridge to create an event pattern.
- ☒ **Custom pattern (JSON editor)**
Write an event pattern in JSON.

Event pattern
Write an event pattern in JSON. You can test the event pattern against the sample event. You can also go to pre-defined pattern.

Prefix matching [Insert](#) ☐ Content-based filter syntax

```
1 {
2   "source": ["aws:ec2"],
3   "detail-type": ["AWS API Call via CloudTrail"],
4   "detail": {
5     "eventSource": ["ec2.amazonaws.com"],
6     "eventName": ["AuthorizeSecurityGroupIngress", "ModifyNetworkInterfaceAttributes"]
7   }
8 }
```

☒ JSON is valid

[Copy](#) [Prettify](#) [Event pattern form](#) [Test pattern](#)

[Cancel](#) [Previous](#) [Next](#)

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22. Configure Target

aws [Search] [Option+S] Asia Pacific (Mumbai) 80%

Dashboard [New](#)

▼ Developer resources

- Learn
- Sandbox
- Quick starts

▼ Buses

- Event buses
- [Rules](#)
- Global endpoints
- Archives
- Replays

▼ Pipes

- Pipes

▼ Scheduler

- Schedules
- Schedule groups

▼ Integration

- Partner event sources
- API destinations
- Connections [Updated](#)

▼ Schema registry

- Schemas

Documentation [↗](#)

Step 1: Event detail

Step 2: Build event pattern

Step 3: **Select target(s)**

Step 4 - optional: Configure tags

Step 5: Review and create

Select target(s)

Permissions

Note: When using the EventBridge console, EventBridge will automatically configure the proper permissions for the selected targets. If you're using the AWS CLI, SDK, or CloudFormation, you'll need to configure the proper permissions.

Target 1

Target types

Select an EventBridge event bus, EventBridge API destination (SaaS partner), or another AWS service as a target.

- ☐ EventBridge event bus
- ☐ EventBridge API destination
- ☒ AWS service

Select a target [Info](#)

Select target(s) to invoke when an event matches your event pattern or when schedule is triggered (limit of 5 targets per rule)

Target location

☒ Target in this account ☐ Target in another AWS account

Topic

Permissions

☒ Use execution role (recommended)

Execution role

EventBridge needs permission to send events to the target specified above. By continuing, you are allowing us to do so. [EventBridge and AWS Identity and Access Management](#)

☒ Create a new role for this specific resource ☐ Use existing role

Role name

► **Additional settings**

[Add another target](#) [Cancel](#) [Skip to Review and create](#) [Previous](#) [Next](#)

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23. Rule will be successfully created

aws Search [Option+S]

Amazon EventBridge

Dashboard [New](#)

▼ Developer resources

- Learn
- Sandbox
- Quick starts

▼ Buses

- Event buses
- [Rules](#)
- Global endpoints
- Archives
- Replays

▼ Pipes

- Pipes

▼ Scheduler

- Schedules
- Schedule groups

▼ Integration

- Partner event sources
- API destinations
- Connections [Updated](#)

▼ Schema registry

- Schemas

Documentation [↗](#)

Rule MonitorSecurityGroup was created successfully

Rules

A rule watches for specific types of events. When a matching event occurs, the event is routed to the targets associated with the rule. A rule can be associated with one or more targets.

Select event bus

Event bus

Select or enter event bus name

default

Rules (1)

[Find rules](#) [Any status](#) [Delete](#) [Enable](#) [Edit](#) [CloudFormation Template](#) [Create rule](#)

☐	Name	Status	Type	ARN	Description
☐	MonitorSecurityGroup	Enabled	Standard	arn:aws:events:ap-south-1:619071313125:rule/MonitorSecurityGroup	Monitoring rule for security groups

24. Navigate to EC2 instances and create an instance

aws

vp

Asia Pacific (Mumbai)

Shravasti

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-015fe3aaafa79d88c)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

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Search

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EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

< 1 >

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	testserver	i-015fe3aaafa79d88c	Running	t2.micro	Initializing	View alarms +	ap-south-1b

Select an instance

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25. Edit inbound rules for EC2 instance

The screenshot displays the AWS Management Console interface for the EC2 service. The left-hand navigation pane shows the 'Instances' section under the 'EC2' category. The main content area is titled 'Instances (1/1)' and shows a table with one instance, 'testserver', with ID 'i-015fe3aaafa79d88c'. The instance is in a 'Running' state. Below the table, the details for the instance are shown, including the 'Security groups' section which lists 'sg-0481f727215ecd328 (launch-wizard)'. The 'Inbound rules' section shows a single rule with ID 'sgr-03bfd2b9fd76ec598' allowing TCP traffic on port 22 from 0.0.0.0/0. The 'Outbound rules' section is also visible but empty.

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-03bfd2b9fd76ec598	22	TCP	0.0.0.0/0

26. Edit inbound rules for EC2 instance

aws [Search] [Option+S] Asia Pacific (Mumbai) Shravasti

EC2 > Security Groups > sg-0481f727215ecd328 - launch-wizard

sg-0481f727215ecd328 - launch-wizard

Details

Security group name launch-wizard	Security group ID sg-0481f727215ecd328	Description launch-wizard created 2025-03-27T12:12:44.088Z	VPC ID vpc-0dbf0938bdb026f6b
Owner 619071313125	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (1)

Search

☐	Name	Security group rule ID	IP version	Type	Protocol
☐	-	sgr-03bfd2b9fd76ec598	IPv4	SSH	TCP

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27. Add two rules, HTTPS and All TCP

aws

Search

[Option+S]

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EC2 > Security Groups > sg-0481f727215ecd328 - launch-wizard > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-03bfd2b9fd76ec598	SSH	TCP	22	Cu... 0.0.0.0/0		Delete
-	HTTP	TCP	80	An... 0.0.0.0/0		Delete
-	All TCP	TCP	0 - 65535	An... 0.0.0.0/0		Delete

Add rule

Cancel Preview changes Save rules

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aws

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EC2 > Security Groups > sg-0481f727215ecd328 - launch-wizard

EC2

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Inbound security group rules successfully modified on security group (sg-0481f727215ecd328 | launch-wizard)

Details

sg-0481f727215ecd328 - launch-wizard

Actions

Details

Security group name launch-wizard	Security group ID sg-0481f727215ecd328	Description launch-wizard created 2025-03-27T12:12:44.088Z	VPC ID vpc-0dbf0938bdb026f6b
Owner 619071313125	Inbound rules count 3 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

Inbound rules (3)

Manage tags Edit inbound rules

1

Name	Security group rule ID	IP version	Type	Protocol
-	sgr-06e5753baf184a901	IPv4	HTTP	TCP
-	sgr-096a0c278dc053a0d	IPv4	All TCP	TCP
-	sgr-03bfd2b9fd76ec598	IPv4	SSH	TCP

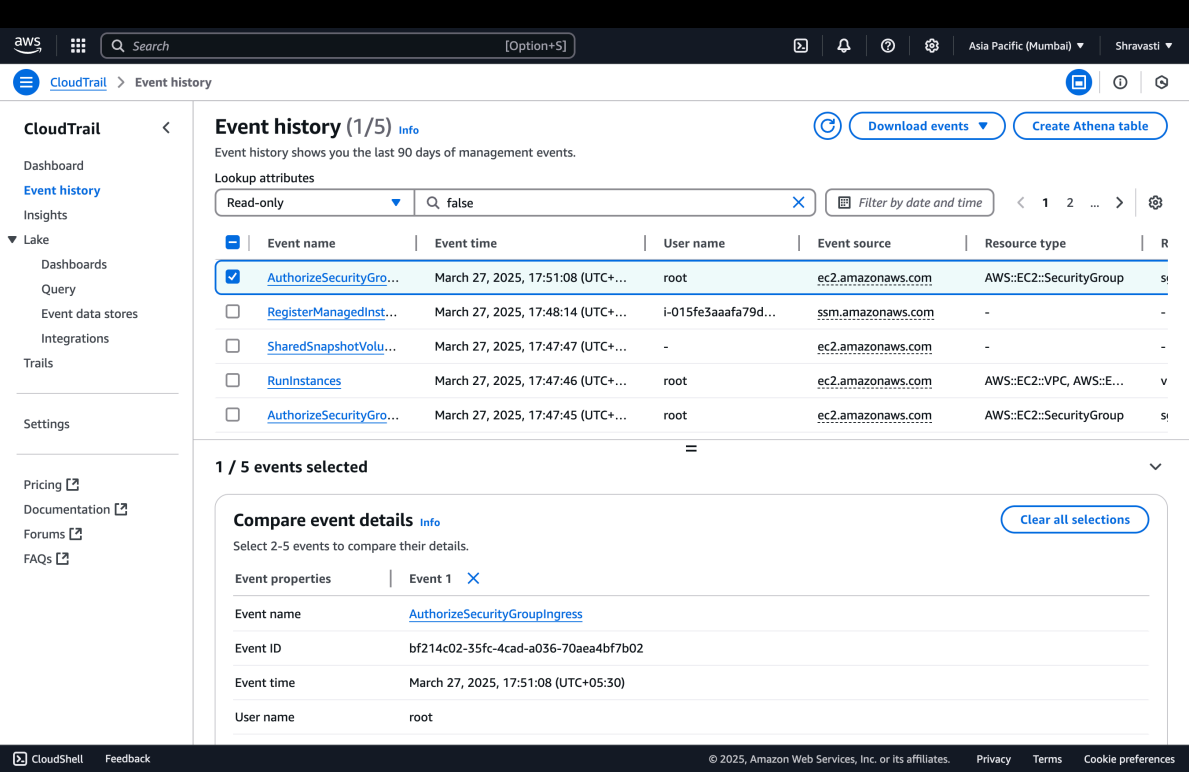
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lab 10 - monitoring and alerting with cloud trail and cloud watch

26

28. Navigate to Cloud Trail, confirm creation of event



The screenshot displays the AWS CloudTrail 'Event history' page. The left sidebar shows the navigation menu with 'Event history' selected. The main content area shows a table of events. The first event, 'AuthorizeSecurityGroupIngress', is selected. Below the table, the 'Compare event details' section shows the properties of the selected event.

Event name	Event time	User name	Event source	Resource type	R
☒ AuthorizeSecurityGroupIngress	March 27, 2025, 17:51:08 (UTC+05:30)	root	ec2.amazonaws.com	AWS::EC2::SecurityGroup	si
☐ RegisterManagedInstance	March 27, 2025, 17:48:14 (UTC+05:30)	i-015fe3aaafa79d...	ssm.amazonaws.com	-	-
☐ SharedSnapshotVolume	March 27, 2025, 17:47:47 (UTC+05:30)	-	ec2.amazonaws.com	-	-
☐ RunInstances	March 27, 2025, 17:47:46 (UTC+05:30)	root	ec2.amazonaws.com	AWS::EC2::VPC, AWS::E...	v
☐ AuthorizeSecurityGroupIngress	March 27, 2025, 17:47:45 (UTC+05:30)	root	ec2.amazonaws.com	AWS::EC2::SecurityGroup	si

1 / 5 events selected

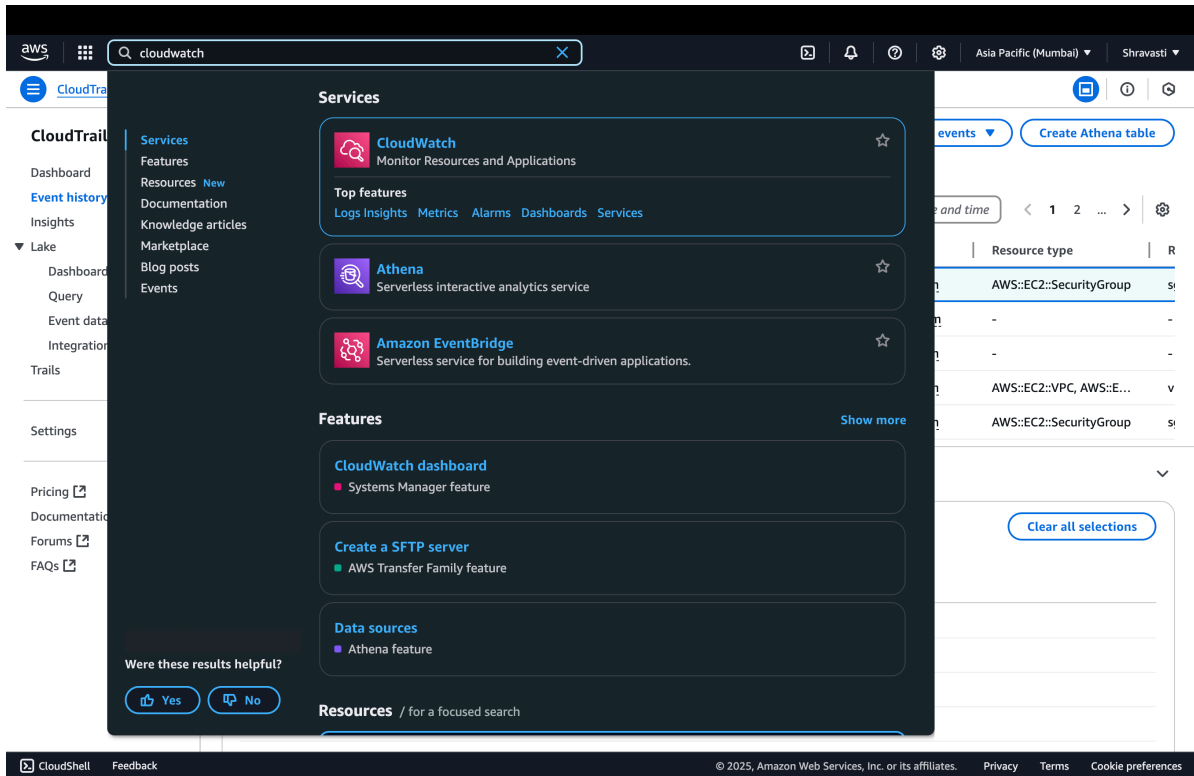
Compare event details [Info](#) [Clear all selections](#)

Select 2-5 events to compare their details.

Event properties | Event 1 [X](#)

Event name	AuthorizeSecurityGroupIngress
Event ID	bf214c02-35fc-4cad-a036-70aea4bf7b02
Event time	March 27, 2025, 17:51:08 (UTC+05:30)
User name	root

29. Navigate to cloud watch



30. Create a log group

aws

Search

[Option+S]

Asia Pacific (Mumbai)

Shravasti

CloudWatch

Log groups

Create log group

CloudWatch

Favorites and recents

Dashboards

Alarms

Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Contributor Insights

Metrics

X-Ray traces

Events

Application Signals

Network Monitoring

Insights

Settings

Getting Started

What's new

Create log group

Log group details

CloudWatch Logs offers two log classes: Standard and Infrequent Access. [Learn more about the features offered by each log class.](#)

Log group name

aws-cloudtrail-logs

Retention setting

Never expire

Log class

Standard

KMS key ARN - optional

Tags

A tag is a label that you assign to an Amazon Web Services resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your Amazon Web Services costs.

No tags are associated with this log group.

Add new tag

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Log groups

CloudWatch

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Dashboards

Alarms

Logs

Log groups

Log Anomalies

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Insights

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Getting Started

What's new

Log group "aws-cloudtrail-logs" has been created.

Log groups (7)

By default, we only load up to 10000 log groups.

Filter log groups or try prefix search

Exact match

1

Log group

Log class

Anomaly d...

Data...

Sens...

Retenti...

☐

[/aws/lambda/GetStartedLambdaProxyIntegration](#)

Standard

[Configure](#)

-

-

Never exp...

☐

[/aws/lambda/arithmetic_operation_mumbai](#)

Standard

[Configure](#)

-

-

Never exp...

☐

[/aws/lambda/my-api-function](#)

Standard

[Configure](#)

-

-

Never exp...

☐

[/aws/lambda/my-function](#)

Standard

[Configure](#)

-

-

Never exp...

☐

[/aws/lambda/palindrome](#)

Standard

[Configure](#)

-

-

Never exp...

☐

[RDSOSMetrics](#)

Standard

[Configure](#)

-

-

1 month

☐

[aws-cloudtrail-logs](#)

Standard

[Configure](#)

-

-

Never exp...

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29

31. Create a metric filter for the log group

The screenshot shows the AWS CloudWatch console interface. At the top, a green notification bar indicates that the log group "aws-cloudtrail-logs" has been created. The main section is titled "Log groups (1/7)" and includes a search bar and a table of log groups. The table has columns for "Log group", "Data source", "Sensitivity", and "Retention". The "aws-cloudtrail-logs" log group is selected, and the "Actions" dropdown menu is open, showing options like "Delete log group(s)", "Edit retention setting(s)", "Create metric filter", "Create contributor insights rules", "Create data protection policy", "Anomaly detection", "Subscription filters", "Export data to Amazon S3", and "View all exports to Amazon S3". The "Create metric filter" option is highlighted.

Log group	Data source	Sensitivity	Retention
☐ /aws/lambda/GetStartedLambdaProxyIntegration	-	-	Never exp...
☐ /aws/lambda/arithmetic_operation_mumbai	-	-	Never exp...
☐ /aws/lambda/my-api-function	-	-	Never exp...
☐ /aws/lambda/my-function	-	-	Never exp...
☐ /aws/lambda/palindrome	-	-	Never exp...
☐ RDSOSMetrics	Standard	Configure	1 month
☒ aws-cloudtrail-logs	Standard	Configure	Never exp...

32. Create a filter pattern with the following name

aws Search [Option+S]

CloudWatch > Log groups > aws-cloudtrail-logs > Create metric filter

Log group "aws-cloudtrail-logs" has been created.

Step 1 Define pattern

Step 2 Assign metric

Step 3 Review and create

Define pattern

Create filter pattern

You can use metric filters to monitor events in a log group as they are sent to CloudWatch Logs. You can monitor and count specific terms or extract values from log events and associate the results with a metric. [Learn more about pattern syntax.](#)

Filter pattern

Specify the terms or pattern to match in your log events to create metrics.

Q {{\$.eventName=ConsoleLogin}}&&(\$.errorMessage="FailedAuthentication")

☐ Enable metric filter on transformed logs

When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Test pattern

Select log data to test

Custom log data

Log event messages

Type log data to test with your Filter Pattern. Please use line breaks to separate log events.

```
[83078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Running Start Crawl for Crawler TestCrawler2
[83078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Classification complete, writing results to database mygluedatabase
[83078518-fcc1-4d30-9573-8b9737671438] INFO : Crawler configured with SchemaChangePolicy
```

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33. Give the filter name as ConsoleLoginErrors

Assign metric

Create filter name

Log events that match the pattern you define are recorded to the metric that you specify. You can graph the metric and set alarms to notify you.

Filter name

ConsoleLoginErrors

Filter pattern

{{\$.eventName=ConsoleLogin}}&&(\$.errorMessage="FailedAuthentication")

☐ Enable metric filter on transformed logs

When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.

Metric details

Metric namespace

Namespaces let you group similar metrics. [Learn more](#)

CloudTrailMetrics

☐ Create new

Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.

Metric name

Metric name identifies this metric, and must be unique within the namespace. [Learn more](#)

ConsoleLoginFailureCount

Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar(\$), and space().

Metric value

Metric value is the value published to the metric name when a Filter Pattern match occurs.

1

Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers (\$1, \$2, etc.), or named field identifiers (e.g. \$requestSize for delimited filter pattern or \$.status for JSON-based filter pattern - dollar (\$) or dollar dot (\$) followed by alphanumeric and/or underscore () characters).

Default value – optional

The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. [Learn more](#)

Enter default value

Unit – optional

Select a unit ▼

34. Review and create the metric filter

- Step 3
Review and create

Review and create

Step 1: Pattern

Edit

Create filter pattern

Filter pattern

```
{($.eventName=ConsoleLogin)&&($.errorMessage="FailedAuthentication")}
```

Step 2: Metric

Edit

Assign metric

Filter name
ConsoleLoginErrors

Metric namespace
CloudTrailMetrics

Metric value
1

Unit

Metric name
ConsoleLoginFailureCount

Applied on transformed logs

Default value
-

Cancel

[Previous](#)

Create metric filter

35. Confirm creation

aws

Search [Option+S]

CloudWatch > Log groups > aws-cloudtrail-logs

Metric filter "ConsoleLoginErrors" has been created.

aws-cloudtrail-logs

Actions View in Logs Insights Start tailing Search log group

Log group details

Log class Info Standard	Metric filters 1	Data protection -
ARN arn:aws:logs:ap-south-1:619071313125:log-group:aws-cloudtrail-logs:*	Subscription filters 0	Sensitive data count -
Creation time 4 minutes ago	Contributor Insights rules -	Field indexes Configure
Retention Never expire	KMS key ID -	Transformer Configure
Stored bytes -	Anomaly detection Configure	

< Log streams Tags Anomaly detection **Metric filters** Subscription filters Contributor Insights Data protection Field indexes - new >

Metric filters (1)

Edit Delete Create alarm [Create metric filter](#)

< 1 > ⚙

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36. Select the metric filter and click on "create alarm"

The screenshot shows the AWS CloudWatch console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and regional information (Asia Pacific (Mumbai)). Below this, the breadcrumb trail indicates the current location: CloudWatch > Log groups > aws-cloudtrail-logs. The main navigation tabs include Log streams, Tags, Anomaly detection, Metric filters (which is the active tab), Subscription filters, Contributor Insights, Data protection, and Field indexes. The 'Metric filters (1/1)' section shows a single filter named 'ConsoleLoginErrors'. The filter details are as follows:

- Filter pattern:** `{{$.eventName=ConsoleLogin}}&{{$.errorMessage="FailedAuthentication"}}`
- Metric:** [CloudTrailMetrics](#) / [ConsoleLoginFailureCount](#)
- Metric value:** 1
- Default value:** -
- Applied on transformed logs:** -
- Unit:** -
- Dimensions:** -
- Alarms:** None.

At the bottom of the console, there's a footer with 'CloudShell', 'Feedback', and copyright information for Amazon Web Services, Inc. or its affiliates.

37. Add metric name, statistic and period

Specify metric and conditions

The screenshot shows the 'Specify metric and conditions' form in the AWS CloudWatch console. The form is divided into two main sections: 'Metric' and 'Conditions'.

Metric Section:

- Graph:** A line graph showing a blue line representing the metric 'ConsoleLoginFailureCount'. The y-axis is labeled 'No unit' and ranges from 0 to 1. The x-axis shows time from 09:30 to 11:30. A red horizontal line is drawn at the 1 mark on the y-axis.
- Namespace:** CloudTrailMetrics
- Metric name:** ConsoleLoginFailureCount
- Statistic:** Sum
- Period:** 5 minutes

Conditions Section:

- Graph:** A line graph showing a blue line representing the metric 'ConsoleLoginFailureCount'. The y-axis is labeled 'No unit' and ranges from 0 to 1. The x-axis shows time from 09:30 to 11:30. A red horizontal line is drawn at the 1 mark on the y-axis.

38. Specify conditions

Conditions

Threshold type

☒ **Static**
Use a value as a threshold

☐ **Anomaly detection**
Use a band as a threshold

Whenever ConsoleLoginFailureCount is...
Define the alarm condition.

☐ **Greater**
> threshold

☒ **Greater/Equal**
≥ threshold

☐ **Lower/Equal**
≤ threshold

☐ **Lower**
< threshold

than...
Define the threshold value.

Must be a number

► **Additional configuration**

Cancel

Next

39. Configure actions

aws

Search

[Option+S]

Asia Pacific (Mumbai)

Shravasti

CloudWatch

Alarms

Create alarm

Step 1

Specify metric and conditions

Step 2

Configure actions

Step 3

Add name and description

Step 4

Preview and create

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
shravasti563@gmail.com - [View in SNS Console](#)

[Add notification](#)

Lambda action

[Add Lambda action](#)

40. Add name and a description

The screenshot shows the AWS CloudWatch console interface for creating a new alarm. The top navigation bar includes the AWS logo, a search bar, and user information (Asia Pacific (Mumbai) and Shrivasti). The breadcrumb trail indicates the path: CloudWatch > Alarms > Create alarm. On the left, a vertical step indicator shows four steps: Step 1 (Specify metric and conditions), Step 2 (Configure actions), Step 3 (Add name and description, which is the current step and highlighted with a blue circle), and Step 4 (Preview and create). The main content area is titled 'Add name and description'. It contains a form with two sections: 'Name and description' and 'Alarm description - optional'. The 'Name and description' section has a text input field labeled 'Alarm name' with the value 'Failed Login'. The 'Alarm description - optional' section has a text area labeled 'Alarm description' with the value '# This is an H1\n**double asterisks will produce strong character**\nThis is [an example](https://example.com/) inline link.' Below the text area is a character count: 'Up to 1024 characters (0/1024)'. A light blue information box below the text area states: 'Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.' At the bottom right of the form, there are three buttons: 'Cancel', 'Previous', and 'Next' (which is highlighted in orange).

aws [Search] [Option+S]

CloudWatch > Alarms > Create alarm

Step 1
Specify metric and conditions

Step 2
Configure actions

Step 3
Add name and description

Step 4
Preview and create

Add name and description

Name and description

Alarm name

Failed Login

Alarm description - optional [View formatting guidelines](#)

Edit Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Cancel Previous Next

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41. Alarm will be created successfully

aws CloudWatch Alarms

Search [Option+S]

Asia Pacific (Mumbai) Shravasti

CloudWatch <

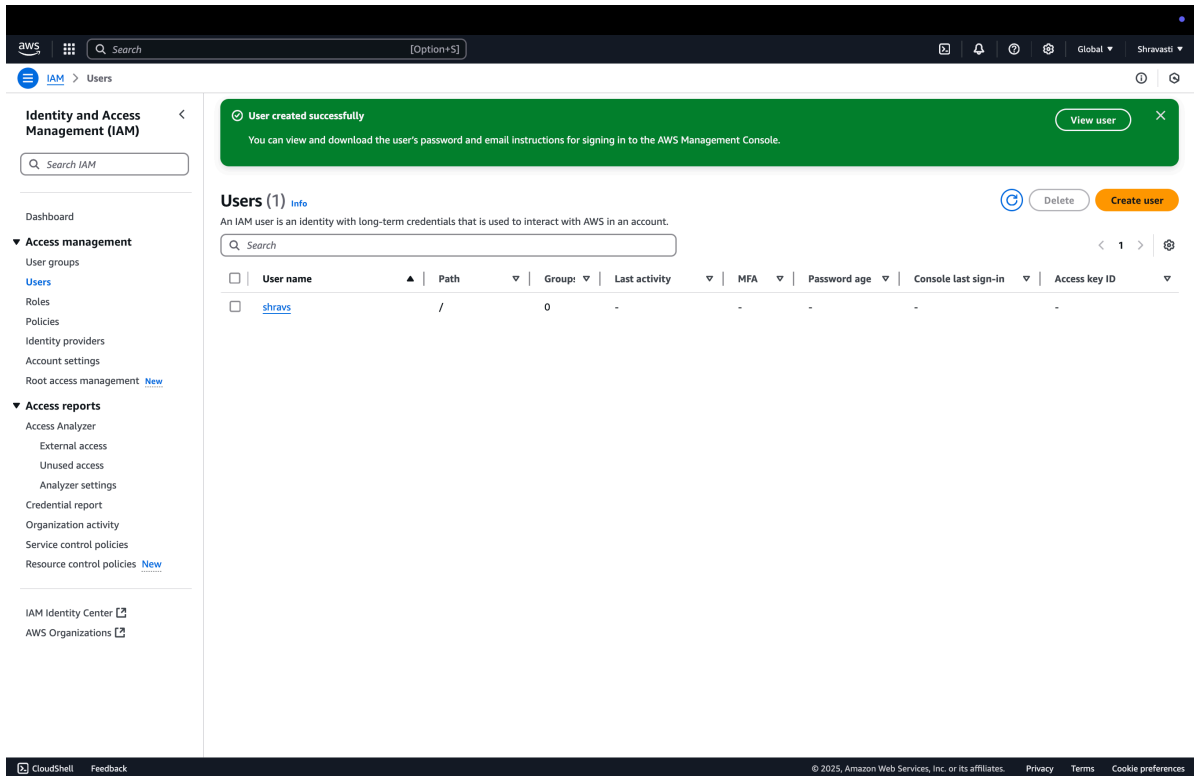
Alarms (1) ☐ Hide Auto Scaling alarms Clear selection Create composite alarm Actions Create alarm

Search Alarm state: Any Alarm type: Any Actions status: Any < 1 > ⚙

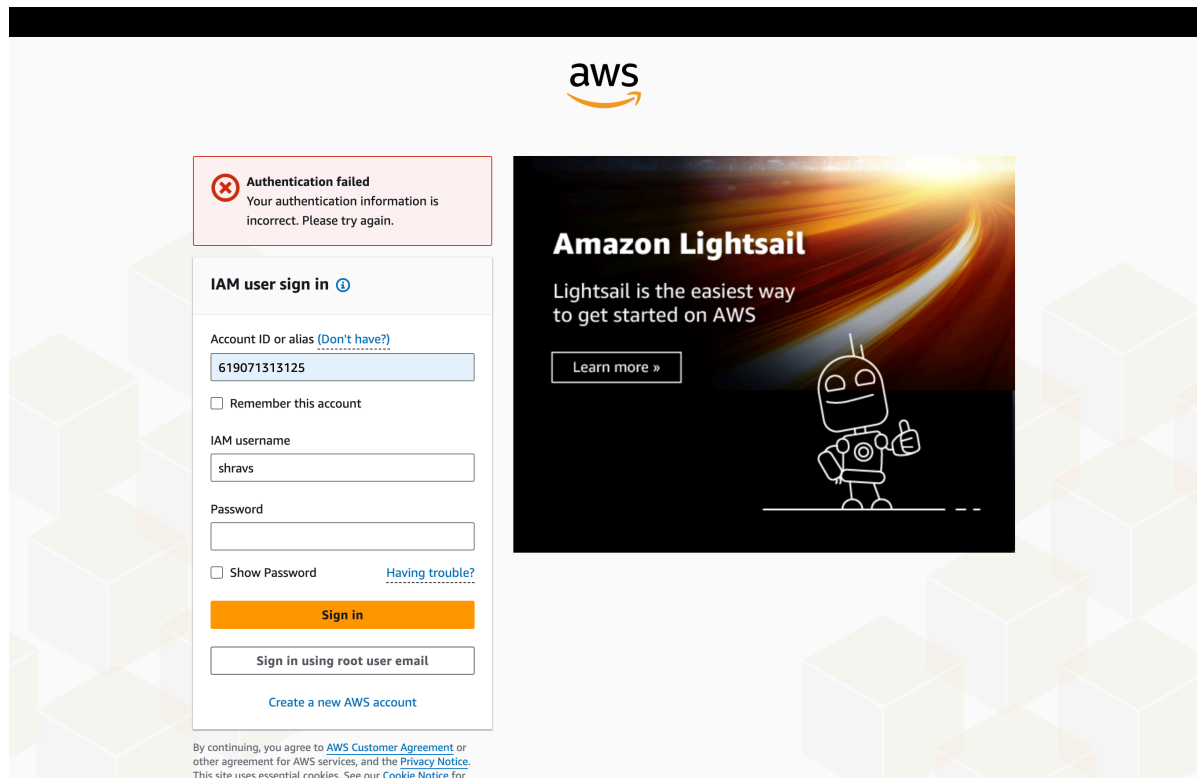
☐	Name	State	Last state update (UTC)	Conditions	Actions
☐	Failed Login	☐ Insufficient data	2025-03-27 12:33:24	ConsoleLoginFailureCount >= 3 for 1 datapoints within 5 minutes	☐

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42. Create an IAM user



43. Log in as this user, authentication will fail



44. Event will be logged

